BCCS: Formal Definition and NP-Hardness Proof

Supplementary Material for ICDM 2026 Submission

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1 Problem Definition

Definition 1 (Budgeted Connected Case Segmentation — BCCS)**.** *Let $G = (V, E)$ be an undirected graph, $L \subseteq E$ a set of* target edges*, and $B \in \mathbb{Z}^+$ a* budget*. A **case assignment** is a partition $\mathcal{P} = \{C_1, \ldots, C_m\}$ of V into pairwise-disjoint, non-empty subsets called* cases*.*

*For each case C_i define the **induced edge set***

$$E(C_i) = \big\{(u, v) \in E : u \in C_i \text{ and } v \in C_i\big\}.$$

*The **coverage** of $\mathcal{P}$ is*

$$\mathrm{Cov}(\mathcal{P}) = \big|\{(u, v) \in L : \exists\, C_i \in \mathcal{P} \text{ with } u, v \in C_i\}\big|.$$

*The **BCCS optimisation problem** is*

$$\max_{\mathcal{P}} \ \mathrm{Cov}(\mathcal{P}) \qquad s.t. \quad |E(C_i)| \le B \ \ \forall\, C_i \in \mathcal{P}.$$

*The **BCCS decision problem** asks: given (G, L, B, τ), does there exist a partition $\mathcal{P}$ with $|E(C_i)| \le B$ for all i and $\mathrm{Cov}(\mathcal{P}) \ge \tau$?*

Interpretation in AML transaction monitoring. In anti-money laundering (AML) systems, G represents the full transaction graph, L is the set of top-k highest-risk edges identified by a GNN scorer, and B limits the number of edges an analyst can review in a single investigation case. BCCS asks how to group transactions into bounded investigation cases to maximise the number of flagged edges that are actually reviewable.

2 NP-Completeness

Theorem 1. *BCCS is NP-complete.*

Proof. **Membership in NP**

Given a partition $\mathcal{P} = \{C_1, \ldots, C_m\}$, we can verify in $O(|V| + |E|)$ time that (i) $|E(C_i)| \le B$ for every i and (ii) $\mathrm{Cov}(\mathcal{P}) \ge \tau$, by building a node-to-case lookup table and scanning all edges once. Hence BCCS $\in$ NP.

NP-Hardness (reduction from Maximum Independent Set)

We reduce from the MAXIMUM INDEPENDENT SET problem (MIS), known to be NP-hard [1].

MIS. Given a graph $H = (V_H, E_H)$ and an integer k, does H contain an independent set of size $\geq k$?

Construction. Given a MIS instance (H, k), build the BCCS instance (G, L, B, τ) as follows.

1. *Add a goal node.* Let g be a fresh node; set $V = V_H \cup \{g\}$.
2. *Edges.* $E = E_H \ \cup \ \{(v, g) : v \in V_H\}$.
3. *Target edges.* $L = \{(v, g) : v \in V_H\}$. Only the edges to g are targets.
4. *Budget and threshold.* $B = \tau = k$.

The construction runs in $O(|V_H| + |E_H|)$ time.

Intuition. Covering a target edge (v, g) requires v and g in the same case. Because g belongs to a single case, all "covered" nodes share that case with g. The non-target edges from E_H between those nodes consume budget, so only independent sets fit within budget k.

Forward direction ($\Rightarrow$). Suppose H has an independent set $S \subseteq V_H$ with $|S| = k$. Define the partition $C_1 = S \cup \{g\}$; every remaining node forms a singleton case.

- *Budget of C_1:* $E(C_1) = E_H(S) \cup \{(v, g) : v \in S\}$. Since S is independent, $E_H(S) = \emptyset$, so $|E(C_1)| = |S| = k = B$. ✓
- *Singletons:* 0 edges each. ✓
- *Coverage:* Each $v \in S$ is co-located with g, giving $\mathrm{Cov} = k = \tau$. ✓

Backward direction ($\Leftarrow$). Suppose a partition $\mathcal{P}$ achieves $\mathrm{Cov}(\mathcal{P}) \geq k$. Let C_g be the case containing g and set $S = C_g \cap V_H$.

A target edge (v, g) is covered if and only if $v \in S$, so $\mathrm{Cov}(\mathcal{P}) = |S| \geq k$.

The induced edges of C_g are

$$E(C_g) = E_H(S) \ \cup \ \{(v, g) : v \in S\},$$

hence $|E(C_g)| = |E_H(S)| + |S|$. The budget constraint gives

$$|E_H(S)| + |S| \ \leq \ B = k.$$

Since $|S| \geq k$, we obtain $|E_H(S)| \leq k - |S| \leq 0$. Therefore $|E_H(S)| = 0$ and $|S| = k$: the set S is an independent set of size k in H. ✓

This completes the reduction. □

3 Consequences

3.1 Inapproximability

The reduction preserves the objective value exactly ($\mathrm{Cov} = |S|$). Any α-approximation for BCCS therefore yields an α-approximation for MIS. Zuckerman [3] showed that MIS cannot be approximated within $|V|^{1-\varepsilon}$ for any $\varepsilon > 0$ unless $\mathrm{P} = \mathrm{NP}$.

Corollary 2. *Unless* $\mathrm{P} = \mathrm{NP}$*, BCCS cannot be approximated within* $|V|^{1-\varepsilon}$ *for any* $\varepsilon > 0$.

This worst-case bound applies to adversarial instances. Real-world AML graphs exhibit temporal locality and natural community structure that make the problem far more tractable in practice.

3.2 Connection to Maximum Coverage

BCCS also relates to the Maximum Coverage Problem (MCP). In settings where each case acts as a "set" of edges it covers, the BCCS optimiser must decide which target edges to cover within the budget, analogous to selecting sets in MCP. Feige [2] showed that MCP cannot be approximated beyond $1 - 1/e \approx 0.632$ unless P = NP.

3.3 Practical Implications

Despite worst-case NP-hardness, our hierarchical WCC + Leiden heuristic (BTCS) achieves **94–97%** of the upper bound (NoBudget WCC coverage) at $k = 1\%$ on the AML100k and AML1M datasets. This suggests that real-world AML transaction graphs possess structure — e.g. natural community boundaries that align with the budget — making the problem tractable in practice even though worst-case instances are intractable.

4 Approximation Guarantees for BTCS

Although BCCS is NP-hard in general, we establish four formal results that characterise the performance of the BTCS algorithm (hierarchical WCC + Leiden) on structured instances. Together, these results explain the strong empirical performance observed across 15 benchmark datasets.

4.1 WCC Coverage as a Computable Upper Bound

Let $G_k = (V_k, E_k)$ denote the subgraph of G induced by the top-k edges (the *top-k subgraph*). Define:

$$\text{WCC-COV}(G, G_k, L) = \big|\{(u, v) \in L : u \text{ and } v \text{ are in the same WCC of } G_k\}\big|.$$

Theorem 3 (WCC Coverage Upper Bound for BTCS)**.** *For all graphs G, top-k subgraphs G_k, target sets L, and budgets $B \geq 1$:*

$$\text{BTCS}(G, G_k, L, B) \;\leq\; \text{WCC-COV}(G, G_k, L).$$

Proof. BTCS partitions nodes according to $\text{WCC}(G_k)$: two nodes u, v are placed in the same case only if they belong to the same WCC (or a Leiden sub-partition thereof). Hence a target edge $(u, v) \in L$ can be covered only if u and v are already co-located in the same WCC of G_k. Therefore BTCS $\leq$ WCC-COV. □

Remark 1. *WCC-COV is computable in $O(|V| + |E_k|)$ and is strictly tighter than $|L|$. Empirically it equals 91.2% and 90.1% of $|L|$ on AML100k and AML1M at $k = 1\%$, so at most 8–9% of target edges are unreachable even without a budget constraint.*

4.2 BTCS Dominates Budget-Capped WCC

The *budget-capped WCC baseline* (WCC-CAP): (i) compute WCC, (ii) treat each component as one case, (iii) apply score-descending truncation to B induced edges. For a WCC component C with $m = |E_G(C)|$ induced edges, $\ell = |L \cap E_G(C)|$ target edges, and $q(C) = \lceil m/B \rceil$:

Theorem 4 (BTCS Multiplicative Gain over WCC-CAP)**.** *Let $f_{\text{cut}}(C) = |L_{\text{cut}}(C)|/\ell$ be the fraction of target edges in C cut between Leiden sub-cases. Then:*

$$\text{cov}_{\text{BTCS}}(C) \;\geq\; \ell\big(1 - f_{\text{cut}}(C)\big) \;\geq\; \text{cov}_{\text{CAP}}(C) \cdot q(C) \cdot \big(1 - f_{\text{cut}}(C)\big).$$

When $f_{\text{cut}}(C) \leq 1 - 1/q(C)$, BTCS covers at least $q(C)$ times more target edges than WCC-CAP in component C.

Proof. A target edge $(u, v) \in L \cap E_G(C)$ is covered by BTCS iff u, v land in the same Leiden subcase, so $\text{cov}_{\text{BTCS}}(C) = \ell(1 - f_{\text{cut}})$. WCC-CAP retains B of m induced edges, covering at most $\ell \cdot B/m = \ell/q(C)$ target edges. Dividing: $\text{cov}_{\text{BTCS}}/\text{cov}_{\text{CAP}} \geq q(C)(1 - f_{\text{cut}})$. □

Empirical validation. At $k = 5\%$ on AML1M, large WCC components have $m \approx 3{,}000$–$20{,}000$ and $B = 100$, giving $q(C) \approx 30$–200. Observed: BTCS = 0.621 vs. WCC-CAP = 0.138, a factor of 4.5×, consistent with Theorem 4.

4.3 Coverage Gap Characterisation

Definition 2 (Cross-WCC Target Edges)**.**

$$\text{cross}(G, G_k, L) = \big|\{(u, v) \in L : u \textit{ and } v \textit{ are in } \text{different } \textit{WCCs of } G_k\}\big|.$$

Theorem 5 (Coverage Gap of BTCS vs. OPT)**.**

$$\text{OPT}(G, L, B) - \text{BTCS}(G, G_k, L, B) \;\leq\; \text{cross}(G, G_k, L) + |L_{\text{cut}}|,$$

where $L_{\text{cut}} = \sum_{C \textit{ over-budget}} L_{\text{cut}}(C)$ are target edges cut by Leiden within over-budget WCC components.

Proof. Target edges covered by OPT but not BTCS fall into two disjoint groups: (i) cross-WCC edges: OPT may merge nodes from different WCCs; BTCS cannot, so at most $\text{cross}(G, G_k, L)$ such edges are missed; (ii) intra-WCC Leiden cuts: for over-budget C, Leiden may separate $(u, v) \in L \cap E_G(C)$; these total $|L_{\text{cut}}|$. The gap is bounded by their sum. □

Corollary 6 (Instance-Specific Approximation Ratio)**.** *Let $\delta = (\text{cross} + |L_{\text{cut}}|)/|L|$. Then:*

$$\text{BTCS}(G, G_k, L, B) \;\geq\; (1 - \delta) \cdot \text{OPT}(G, L, B).$$

In AML graphs with strong temporal locality, δ is empirically small, yielding a near-optimal instance-specific ratio.

4.4 Budget-Feasibility Rate and Near-Optimality

Definition 3 (Budget-Feasible Instance)**.** *Instance (G, G_k, L, B) is* budget-feasible *if every WCC component C of G_k satisfies $|E_G(C)| \leq B$ (no Leiden splitting required).*

Theorem 7 (BTCS Optimality on Budget-Feasible Instances)**.** *On budget-feasible instances:*

$$\text{BTCS}(G, G_k, L, B) = \text{WCC-COV}(G, G_k, L), \qquad L_{\text{cut}} = \emptyset.$$

The gap vs. OPT reduces to $\text{cross}(G, G_k, L)$ alone.

Proof. When all WCCs satisfy $|E_G(C)| \leq B$, BTCS retains each WCC as a single case (no Leiden splitting), covering all intra-WCC target edges. This equals WCC-COV by definition; no intra-WCC cuts occur. □

Proposition 8 (Budget-Feasibility Rate in AML Graphs). *At $k = 1\%$ with $B = 100$, the AML100k dataset has 2,546 cases of which only 38 require splitting (BFR = 98.5%), and AML1M has 22,170 cases of which 448 require splitting (BFR = 98.0%). At $k = 1\%$, BTCS operates in the near-budget-feasible regime, achieving 96.5% and 94.2% of the NoBudget upper bound, respectively. The remaining gap comes entirely from the $\approx 2\%$ of cases requiring Leiden splitting.*

Remark 2 (Coverage Degradation at $k = 5\%$). *As k increases, larger WCC components form: at $k = 5\%$ on AML1M, over 99% of top-k edges belong to components exceeding $B = 100$. Budget-feasibility collapses and WCC-CAP degrades to 13.8% coverage. BTCS's Leiden splitting recovers 62.1% via the multiplicative gain of Theorem 4 ($q(C) \gg 1$ for large components), confirming the $4.5\times$ empirical gain at $k = 5\%$.*

References

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